Predicting Resolution Time for City Service Requests Using Machine Learning

Boston 311 Dataset | 2015 - 2024 | 2.5 M Records

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KEY RESULTS

Best Model: CatBoost GPU (MAE loss) -- MAE = 2.669 days

33.3% improvement over mean baseline | 24.1% over SARIMA

21 configurations across 6 model families | 99 features | 1.8 M records

1. Introduction

City 311 systems receive millions of non-emergency service requests annually, covering issues such as potholes, broken streetlights, graffiti removal, and sanitation. Accurately predicting how long each request will take to resolve can help city agencies allocate resources more efficiently and provide citizens with realistic expectations.

This project builds a machine learning pipeline to predict resolution time for Boston 311 service requests using only information available at the time the request is created. A strict creation-time feature boundary ensures predictions can be made immediately for each incoming request. Prior work by Raj et al. (2021) -- the SWIFT system -- addressed the same domain but forecasted weekly aggregate resolution times using sparse Gaussian CRFs. This project extends the problem to per-request prediction using request-level ML features.

The project evaluates multiple modeling stages: statistical baselines, linear models, gradient boosting with default and tuned hyperparameters, robust loss functions, ensemble methods, and comparisons against time-series (ARIMA/SARIMA) approaches.

2. Dataset

Source: City of Boston Open Data Portal (data.boston.gov)
Time range: 2015-2024 (10 years)
Raw records: 2,550,536
After filtering: 1,823,856 (closed cases, 0-90 day cap, excluding administrative closures)

Data Filtering

Stage	Records	Dropped
Raw data	2,550,536	--
Closed cases only	2,348,040	202,496
Valid dates	2,348,038	2
Non-negative resolution	2,347,246	792
90-day cap	2,217,370	129,876
Remove admin closures	1,823,856	393,514

Temporal Split

Split	Years	Records	Percentage
Train	2015-2021	1,190,355	65.3%
Validation	2022	224,371	12.3%
Test	2023-2024	409,130	22.4%

For advanced model selection (Step 4 onward), the training block is further split internally: 2015-2020 serves as train-proper for fitting candidate models, and 2021 is held out as a tune fold for all selection decisions (loss function sweeps, Optuna hyperparameter tuning, ensemble weight optimization, and hurdle threshold selection). The validation set (2022) is used only for early-stopping callbacks, never for model selection, preventing validation leakage. After selection, final models are retrained on the full training block (2015-2021) with 2022 for early stopping before test evaluation.

Target Distribution

Mean resolution: 4.50 days
Median resolution: 0.47 days
Skewness: 4.23
64% of requests resolve within 1 day, creating a heavily right-skewed distribution.

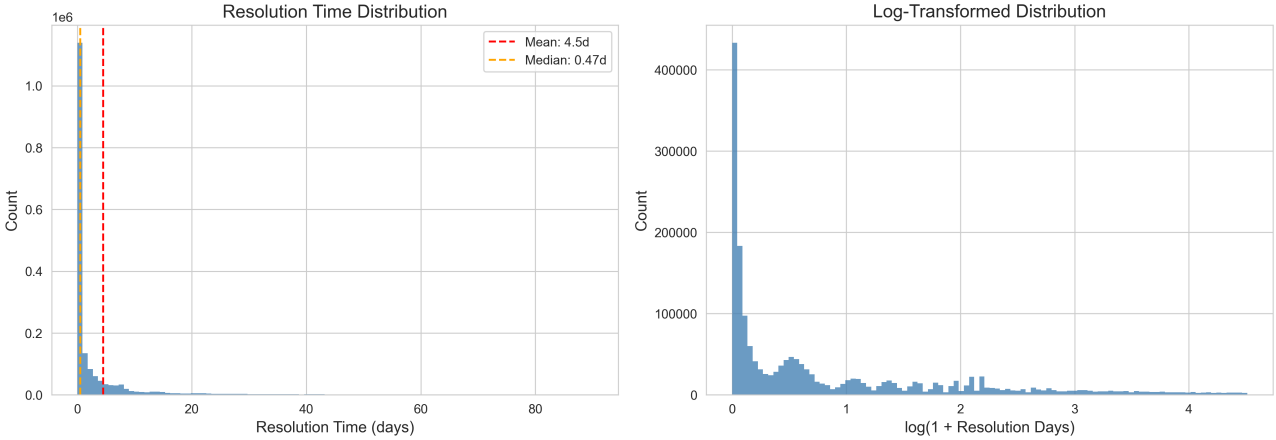

Figure 1: Resolution time distribution

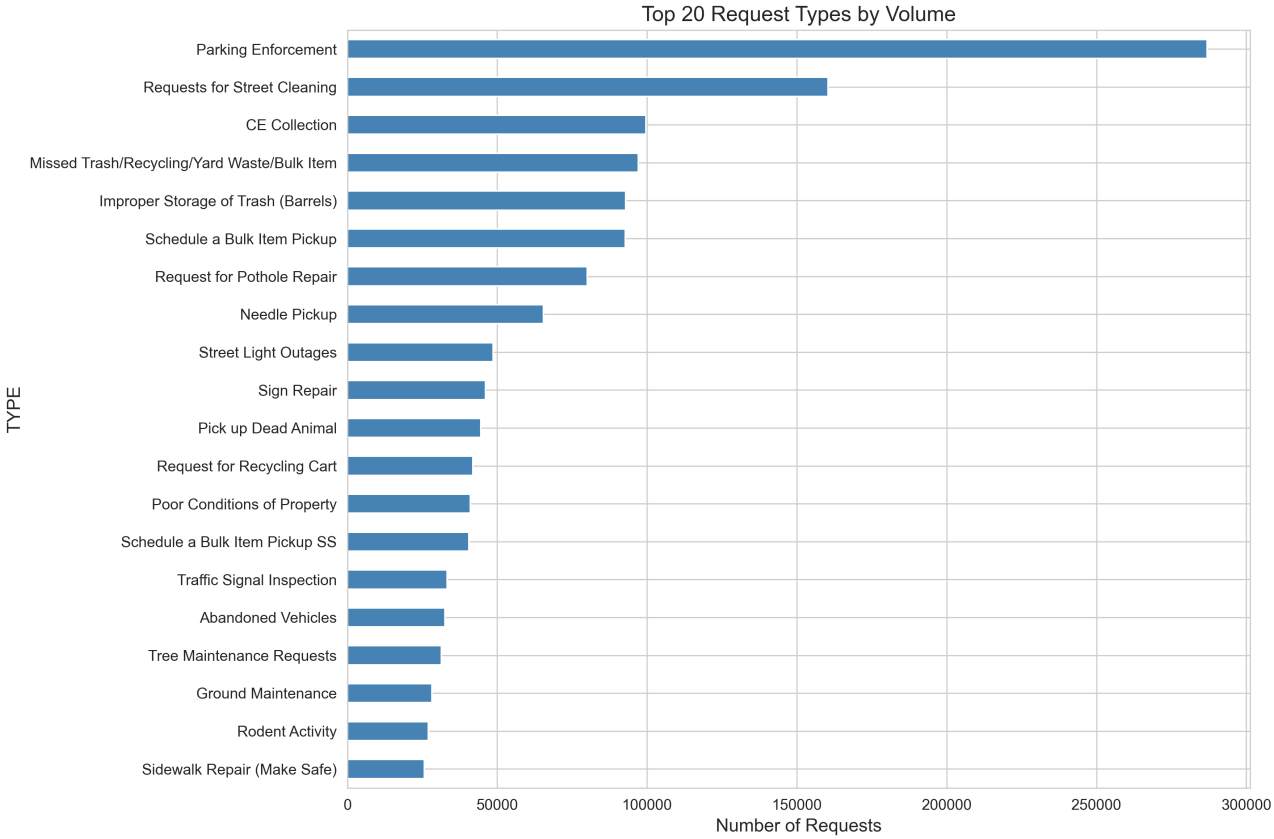

Figure 2: Top request categories

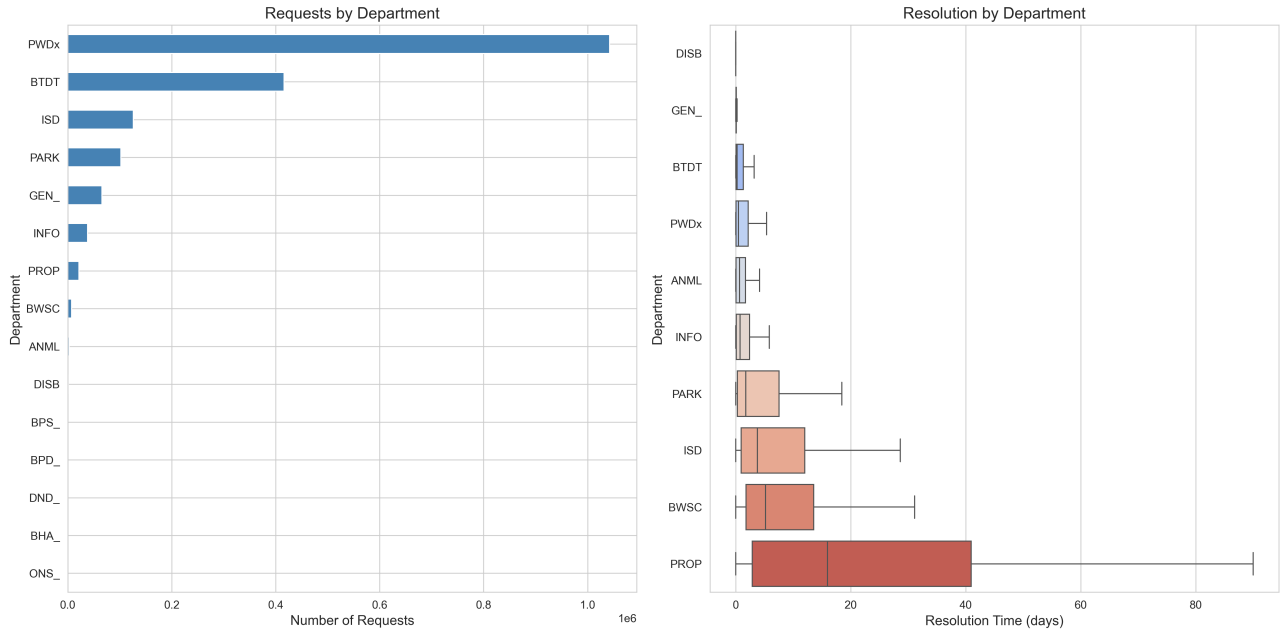

Figure 3: Department analysis

3. Feature Engineering (99 Features)

All features respect the creation-time feature boundary. Historical aggregates and velocity features use only training data or appropriately lagged values to prevent data leakage.

Category	Count	Key Examples
Temporal	27	hour, day_of_week, year, cyclical, time-of-day
Categorical (enc.)	26	Target + frequency encoding
Geographic	5	Lat/Lon, zipcode, dist. from center
Historical aggs.	15	Mean/median/std by TYPE, REASON
Workload proxies	2	Prev day/week request counts
District backlog	2	Open cases (1-day lag)
Dept. velocity	6	Rolling 7/14/30-day averages
Interaction	8	type x dept, velocity x backlog
SLA baseline	6	P25/P75/IQR per TYPE, REASON
Velocity deviation	2	Current vs historical speed

Leakage Prevention

- District backlog: Uses 1-day lag on case closures
- Workload proxies: Previous day/week counts only
- Historical features: Computed from training data only
- Target encoding: Fitted on training set, applied to val/test
- Velocity features: Rolling averages from training period

4. Full Model Progression

All models were trained on the same data pipeline (99 features, 90-day cap, temporal split) and evaluated on the held-out test set (2023-2024, 409,130 records). This ensures a fair, apples-to-apples comparison across all model families.

Steps 1-3 (baselines, linear models, default gradient boosting) train on the full training block (2015-2021) with 2022 for early stopping, since no hyperparameter selection is involved. Step 4 (advanced LightGBM) uses the nested train-proper / tune fold split described in Section 2 to make all selection decisions without touching the validation set.

Step 1: Statistical Baselines

Model	MAE (days)	MedAE	RMSE	R2	Improv.
Mean baseline	4.002	1.431	10.89	-0.039	0.0%
Median baseline	3.577	0.537	11.10	-0.080	10.6%

Step 2: Linear Models

Model	MAE (days)	MedAE	RMSE	R2	Improv.
Linear Regression	2.994	0.302	9.53	0.204	25.2%
Ridge	2.994	0.302	9.53	0.204	25.2%
Lasso	2.987	0.299	9.49	0.210	25.4%
Decision Tree	2.933	0.257	9.25	0.250	26.7%

Even simple linear models achieve ~25% improvement, showing the features carry strong signal.

Step 3: Gradient Boosting (Default Params)

Model	MAE (days)	MedAE	RMSE	R2	Improv.
Random Forest	2.831	0.265	9.00	0.290	29.3%
XGBoost (CUDA)	2.779	0.261	8.91	0.304	30.6%
CatBoost (GPU)	2.757	0.268	8.90	0.305	31.1%

Gradient boosting models gain 4-6% over linear models. CatBoost slightly leads.

Step 4: Advanced LightGBM (Robust Loss Functions)

Model	MAE (days)	MedAE	RMSE	R2	Improv.
LGB Tuned (Optuna)	2.696	0.256	8.73	0.332	32.6%
LGB Huber	2.705	0.261	8.74	0.330	32.4%
LGB Quantile	2.737	0.219	8.79	0.322	31.6%
LGB Fair	2.749	0.262	8.89	0.308	31.3%
Hurdle Soft (0.5d)	2.754	0.256	8.97	0.294	31.2%
LGB L2 + Monotonic	2.765	0.270	8.91	0.305	30.9%
Hurdle Hard (0.5d)	2.812	0.216	8.63	0.348	29.7%
Blended Ensemble*	2.749	0.262	8.89	0.308	31.3%
LGB Tweedie	3.106	0.297	8.45	0.374	22.4%

LGB Tuned (Optuna, Fair loss) achieves the best single-model LightGBM MAE. The SLSQP-optimized blended ensemble collapsed to 100% weight on lgb_fair, so the 'blended ensemble' row is effectively equivalent to LGB Fair. The meta-ensemble (blend + hurdle) similarly assigned 0% hurdle weight, making it also identical to LGB Fair.

5. Literature Comparison (ARIMA/SARIMA)

SWIFT (Raj et al., 2021) compared sparse GCRFs against Linear Regression, ARIMA, and SARIMA baselines for weekly aggregate forecasting. To situate the per-request approach against similar time-series methods, ARIMA and SARIMA models are trained on weekly aggregated resolution times and each test request is assigned its period forecast, enabling a direct per-request MAE comparison.

Model	MAE (days)	RMSE	MedAE
Best ML (LGB Tuned)	2.696	8.73	0.256
Linear Regression	2.994	9.53	0.302
SARIMA(1,1,1)(1,1,1,52)	3.551	11.15	0.387
ARIMA(2,1,2) weekly	3.554	11.14	0.413

ML advantage over SARIMA: 24.1% lower MAE (LGB Tuned vs SARIMA). The best overall model (CatBoost GPU, MAE=2.669d) achieves 24.8% improvement over SARIMA. Per-request ML prediction with creation-time features substantially outperforms aggregate time-series forecasting.

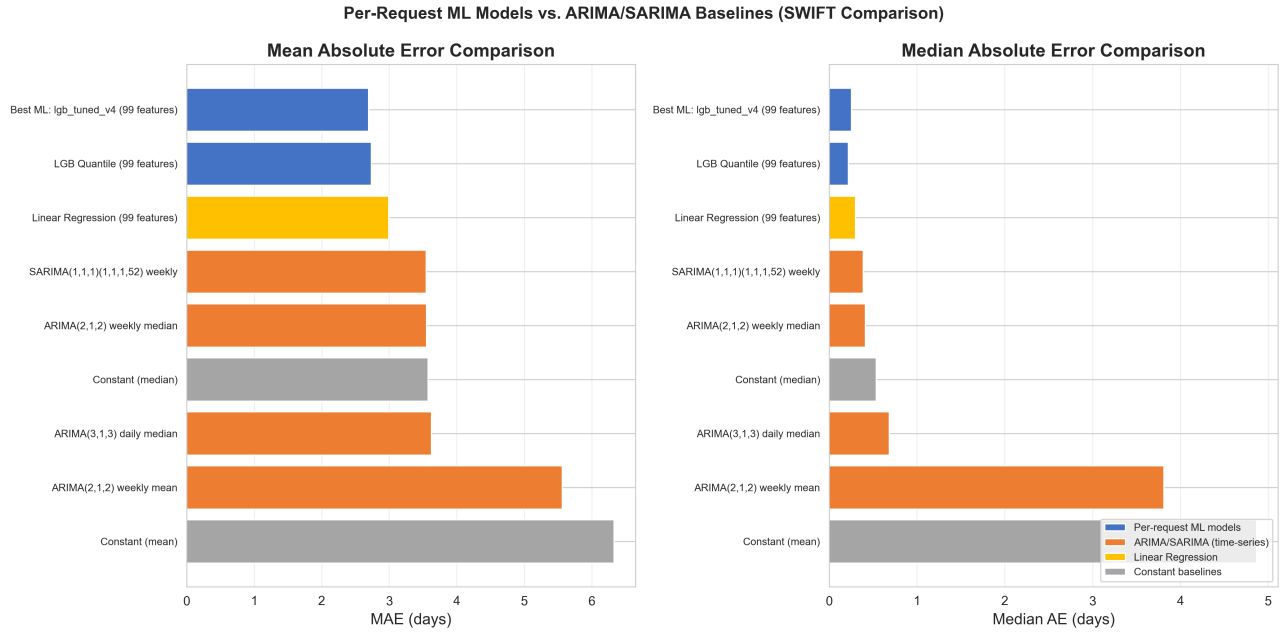

Figure 4: ML vs ARIMA/SARIMA comparison

6. Late-Stage Improvement Experiments

After reaching the best LightGBM (MAE~2.696 days), this work explored whether alternative algorithms or techniques could push performance further. A frozen Huber/Quantile blend baseline (~2.704 days) served as the comparison target for each experiment. CatBoost and XGBoost experiments used GPU acceleration; LightGBM-based experiments ran on CPU.

Experiment	MAE (days)	Beats baseline?	Device
CatBoost MAE loss	2.669	Yes (-1.3%)	GPU
XGBoost abs. error	2.695	Yes (-0.3%)	CUDA
Recency weighting	2.706	No	CPU
Deeper Optuna (50)	2.730	No	CPU
Recency (decay=2.0)	2.728	No	CPU

CatBoost with MAE loss achieves the best overall MAE of 2.669 days, beating the blended ensemble baseline (~2.704 days). Most experiments fail to beat the baseline, confirming the performance plateau is real and data-driven, not a tuning limitation.

Walk-Forward Validation

Walk-forward validation retrains a LightGBM Huber model (delta=0.5, default hyperparameters) on expanding windows to assess temporal stability. Each year's model is trained on all preceding years without a separate validation set.

Test Year	MAE (days)	R-squared	Train Size
2022	2.426	0.310	1,190,355
2023	2.653	0.359	1,414,726
2024	2.662	0.330	1,617,961

Performance is stable across years with no evidence of catastrophic temporal drift. Note: these results use a standard LightGBM Huber model, not the final tuned model or CatBoost, so they represent a conservative stability check.

7. Error Analysis

Error analysis and SHAP explanations are generated from the best LightGBM model (lgb_tuned_v4, MAE=2.696d) rather than the CatBoost improvement experiment, because the LightGBM pipeline produces the full suite of diagnostic artifacts (SHAP values, error breakdowns, predicted-vs-actual plots) during Step 7.

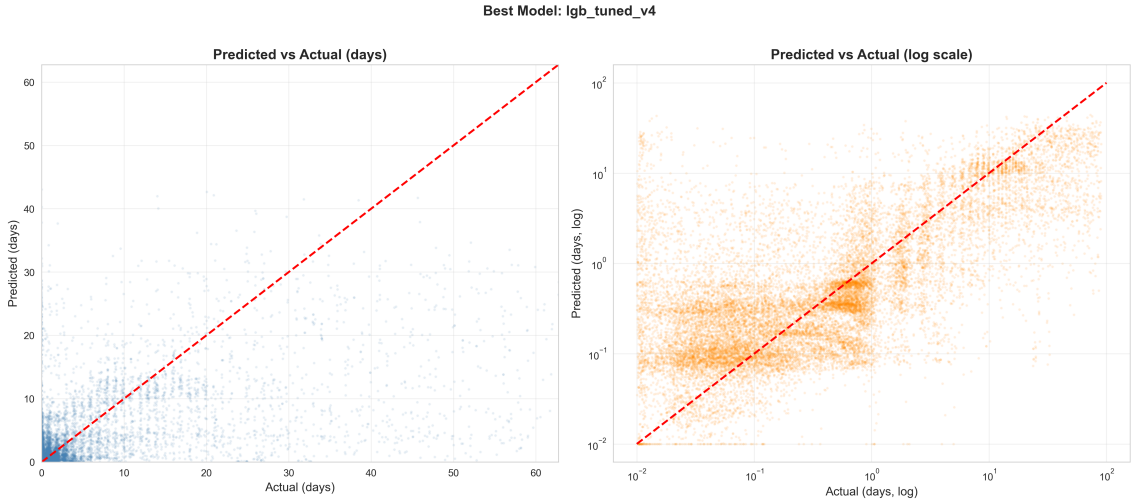

Figure 5: Predicted vs actual (lgb_tuned_v4)

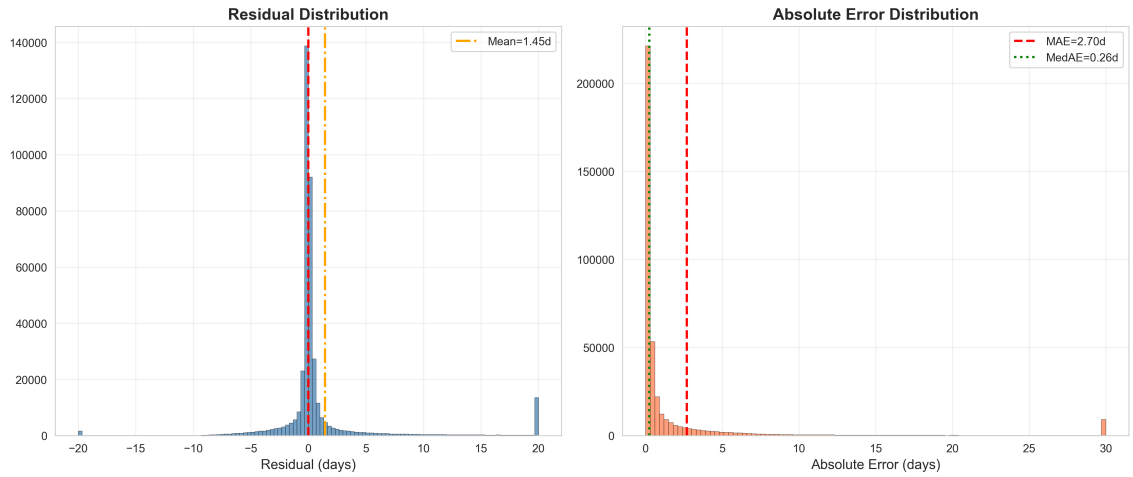

Figure 6: Error distribution (lgb_tuned_v4)

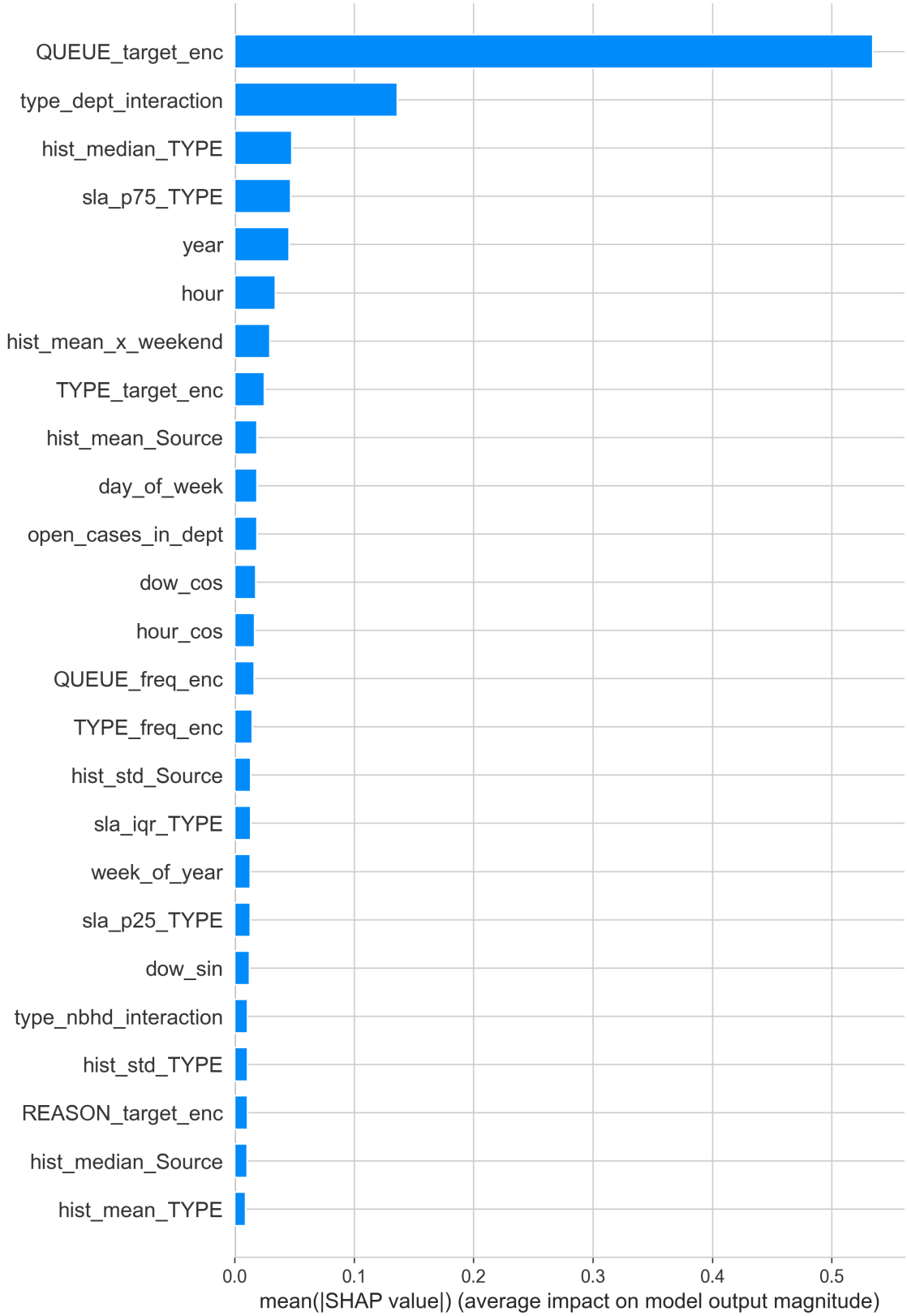

Figure 7: SHAP feature importance (lgb_tuned_v4)

8. Why Further Improvement is Difficult

1. Heavy-tailed target distribution: 64% of requests resolve within 1 day, but the remaining 36% span up to 90 days. The long tail dominates MAE/RMSE.

2. Inherent operational noise: Resolution time depends on staffing, weather, equipment availability, and political priorities -- none of which are in the data.

3. Same-category variance: Requests of the same TYPE and Department can resolve in hours or weeks depending on severity, which is not captured in structured fields.

4. Log retransformation bias: Training on log1p(target) introduces systematic bias when converting predictions back to days.

5. Metric sensitivity: R-squared is computed against the mean, so the extremely skewed distribution makes it structurally low even with good predictions.

9. Development History

The project was developed iteratively. The final repository contains a single active pipeline producing the authoritative results reported throughout this document. Key milestones during development:

Stage	Features	What Changed	Test MAE
Early	~65	Initial features (had data-leakage issues)	N/A*
Intermediate	~83	Leakage-free rewrites + interaction features	2.726
Final pipeline	99	SLA baselines, velocity deviation, Optuna tuning	2.696
+ CatBoost MAE	99	CatBoost GPU with MAE loss (improvement exp.)	2.669

* Early experiments used per-split rolling and backlog functions that had subtle data leakage. These were replaced with leakage-free implementations that compute features across all splits chronologically with appropriate lag constraints.

The final best result -- CatBoost GPU with MAE loss (2.669 days) -- comes from a late-stage improvement experiment (Step 9) using the same 99-feature pipeline.

10. Conclusions

This project demonstrates that machine learning can meaningfully predict Boston 311 service request resolution times, achieving a 33.3% improvement over a naive mean baseline and 24.1% improvement over SARIMA forecasting.

Key contributions:
- Comprehensive model comparison across 21 configurations and 6 model families
- Strict creation-time feature boundary ensuring operational deployability
- 99 engineered features including novel velocity deviation and SLA baseline features
- Honest assessment of the performance plateau and its data-driven causes
- Walk-forward validation (LightGBM Huber) confirming temporal stability
- GPU-accelerated training on NVIDIA RTX 5070 Ti (CUDA/CatBoost GPU)

The best model (CatBoost GPU with MAE loss) achieves:
- MAE: 2.669 days (33.3% improvement over mean baseline)

Walk-forward validation using a LightGBM Huber model shows stable MAE across 2022-2024 (2.43-2.66 days), confirming the feature pipeline generalizes temporally.

The best tuned LightGBM (MAE = 2.696 days) is only 1.0% behind CatBoost, confirming the performance plateau is inherent to the data, not a modeling limitation. Further gains would require richer data sources (free-text descriptions, images, real-time staffing data) rather than algorithmic changes.

References

[1] R. Raj, A. Ramesh, A. Seetharam, and D. DeFazio, "SWIFT: A non-emergency response prediction system using sparse Gaussian Conditional Random Fields," Pervasive Mob. Comput., vol. 71, p. 101317, Feb. 2021.